

TOTALLY INVARIANT SUBVARIETIES OF $\mathbf{P}^n$

GPT 5.6 PRO, COMMUNICATED BY YUJIE LUO AND SHENG MENG

ABSTRACT. Let $f : \mathbf{P}_{\mathbb{C}}^n \rightarrow \mathbf{P}_{\mathbb{C}}^n$ be an endomorphism of degree greater than one. We prove that every irreducible Zariski closed subset V with $f^{-1}(V) = V$ is a linear subspace.

Disclosure. We claim no credit for this work. We are sharing it solely to present a proof generated by GPT 5.6 Pro. We have simplified and verified the proof.

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1. INTRODUCTION

We work over $\mathbb{C}$. Let $f : \mathbf{P}^n \rightarrow \mathbf{P}^n$ be a surjective endomorphism of algebraic degree $q > 1$, i.e., $f^*\mathcal{O}_{\mathbf{P}^n}(1) \simeq \mathcal{O}_{\mathbf{P}^n}(q)$. This is equivalent to $\deg f = q^n > 1$. An irreducible Zariski closed subset $V \subseteq \mathbf{P}^n$ is said to be *totally invariant* if $f^{-1}(V) = V$. The folklore conjecture below asserts the linearity of totally invariant subvarieties.

Linearity Conjecture. *Let $f : \mathbf{P}^n \rightarrow \mathbf{P}^n$ be an endomorphism of degree greater than one. Then any totally invariant irreducible Zariski closed subset is linear.*

We briefly review the historical development of this conjecture.

In dimension two, Fornæss–Sibony obtained early degree bounds for totally invariant curves in $\mathbf{P}^2$ and reduced the nonlinear case to that of a smooth conic [15, Proposition 4.2 and Section 4, p. 212]. Cerveau–Lins Neto ruled out this remaining possibility, thereby establishing the conjecture for plane curves [8, Theorem 2 and Sections 3–4]. Gurjar subsequently proved a ramification theorem that yields the same conclusion as a corollary [17], providing another proof of the plane case; see [6, Remark 5]. In arbitrary dimension, Cerveau–Lins Neto and Beauville proved the conjecture for smooth hypersurfaces [8, 1]. The main difficulties beyond these results concern singular hypersurfaces and positive-dimensional subvarieties of higher codimension, particularly singular ones.

The exceptional set exists [13, Theorem 6.1]. Moreover, for a fixed endomorphism, there are only finitely many irreducible proper totally invariant

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Zariski-closed subsets of $\mathbf{P}^n$ [13, Corollary 6.5]. In 2003, Dinh–Sibony gave the first correct proof of existence in the more general setting of polynomial-like maps [12, Section 3.4]. The projective-space result is recalled in their 2008 paper [13, Section 6]. In 2009, Dinh gave another proof using analytic multiplicative cocycles [11, Theorem 3.1].

In 2004, Briend–Cantat–Shishikura presented an incomplete proof of the linearity conjecture in full generality [2]. Their argument relied on Briend–Duval’s local-degree lemma [3, Section 2]. Dinh pointed out the error in this lemma in November 2006, as recorded in Briend–Duval’s 2009 erratum [4]; the error concerns the counting of multiplicities [13, p. 328]. The erratum proposes a replacement argument after passing to an iterate [4], but does not prove the linearity conjecture. The gap in Briend–Cantat–Shishikura’s proof is explicitly noted by Cantat–Xie [6, Remark 6].

D.-Q. Zhang studied singular totally invariant hypersurfaces in $\mathbf{P}^3$ and reduced the nonlinear case to four possible singular cubic surfaces [24, Theorem 1.1]. Combining a logarithmic-differential estimate with earlier results, including Nakayama–Zhang’s exclusion of the degree-two case [21, arXiv version, Section 5], Horing subsequently proved that every totally invariant prime divisor in $\mathbf{P}^3$ is a hyperplane [19, Corollary 1.2 and its proof]. In arbitrary dimension, Mabed extended this method to prove that every irreducible totally invariant hypersurface in $\mathbf{P}^n$, $n \geq 2$, with at most isolated singularities is a hyperplane [20, Corollary 1.4]. Broustet–Horing’s recent preprint extends the hypersurface case to $\mathbf{P}^4$ and proves, in arbitrary dimension, that every normal totally invariant hypersurface in $\mathbf{P}^n$, $n \geq 2$, is a hyperplane [5, Corollaries 1.5 and 1.3].

In higher codimension, Chen–Meng–Zhong obtained a linearity result for smooth curves in $\mathbf{P}^3$ under an additional equivariance assumption. They proved that every smooth rationally connected projective threefold of Picard number two admitting an int-amplified endomorphism is toric [10, Theorem 1.3]. In particular, this conclusion holds for such threefolds admitting a polarized endomorphism. As a corollary, every blowup-equivariant smooth curve in $\mathbf{P}^3$ for a non-isomorphic surjective endomorphism is a line [10, Corollary 1.6].

In this paper, we prove the following theorem, thereby establishing [Linearity Conjecture](#) in full.

Linearity Theorem. *Let $f : \mathbf{P}^n \rightarrow \mathbf{P}^n$ be an endomorphism of degree greater than one. Let $V \subseteq \mathbf{P}^n$ be an irreducible Zariski closed subset such that $f^{-1}(V) = V$. Then V is a linear subspace.*

Remark 1.3. [Linearity Theorem](#) also holds over any algebraically closed field of characteristic zero. Indeed, descend f and V to a field finitely generated over $\mathbb{Q}$ and embed it into $\mathbb{C}$ [23, Tag 01ZM]; both the hypotheses and linearity can be checked after field extension.

We will use the following setup.

Setup 1.4. We use the following notation.

- $f = [f_0 : \cdots : f_n]$, $g = f|_V$.
- $r = \dim V$, $d = \deg V$, $q = (\deg f)^{1/n}$, $t = r(q - 1) - 1$.
- $\{\ell_i, \tilde{\ell}_i\}_{i=1}^r$, $y_1, \dots, y_d$, $S = S_1 \sqcup \cdots \sqcup S_d$, $(v_x)_{x \in S}$.

We construct them as follows. In the setting of [Linearity Theorem](#), assume for a contradiction that

$$1 \leq r < n, \quad d \geq 2.$$

The f_i are degree- q homogeneous forms, with $q > 1$ and $\deg g = q^r$. Denote the smooth locus of V by V_{reg} . Choose a dense open set $U \subseteq V_{\text{reg}}$ such that $g^{-1}(U) \subseteq V_{\text{reg}}$ and $g^{-1}(U) \rightarrow U$ is étale.

By Bertini's theorem, we can choose a general codimension- r linear subspace $L = \{\ell_1 = \cdots = \ell_r = 0\} \subseteq \mathbf{P}^n$, where the ℓ_i are linear forms, meeting V transversely in

$$V \cap L = \{y_1, \dots, y_d\} \subseteq U.$$

Put

$$\tilde{\ell}_i = \ell_i(f_0, \dots, f_n), \quad S_j = g^{-1}(y_j).$$

The common zero scheme of $\tilde{\ell}_1, \dots, \tilde{\ell}_r$ on V is S , with each S_j consisting of q^r distinct smooth points of V . For each $x \in S$, fix a homogeneous coordinate representative $v_x \in \mathbb{C}^{n+1} \setminus \{0\}$ with $[v_x] = x$. For a homogeneous polynomial P , write $P(x) := P(v_x)$ in the evaluation relations below.

The construction applies after replacing f by any iterate; the geometric data are then chosen for that iterate, while V remains fixed.

Proof of [Linearity Theorem](#). The cases $r = 0$ and $r = n$ are immediate. Otherwise, suppose for a contradiction that $d > 1$ and work in Setup 1.4. Fix $m > 0$ as in Lemma 4.2. Replace f by an iterate such that $q > md$, and choose the remaining data of Setup 1.4 for this iterate. Lemma 4.2 gives a homogeneous polynomial Q of degree m such that $Q|_V \not\equiv 0$ and Q vanishes on $S \setminus S_1$.

Let Z be the zero divisor of $Q|_V$, an effective Cartier divisor since V is integral. Choose $r-1$ general pullbacks $E_i \in |\mathcal{O}_V(q)|$ of hyperplanes through y_2 (none if $r = 1$). Their base locus S_2 is finite, so they meet Z properly, giving an effective zero-cycle. Each of the q^r distinct smooth points of S_2 contributes at least one: the local equations form a system of parameters. The intersection formula (see [16, Sections 2.3–2.4]) gives

$$q^r \leq \deg(Z \cdot E_1 \cdots E_{r-1}) = q^{r-1} dm < q^r,$$

a contradiction. $\square$

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2. A RELATIVE JACOBIAN LEMMA

For an Artinian local ring $(Q, \mathfrak{m}_Q)$, its socle is $\text{Soc}(Q) = \text{Ann}_Q(\mathfrak{m}_Q)$. The Jacobian theorem of Scheja–Storch states that, over a field of characteristic zero, the socle of a zero-dimensional complete intersection

$$Q = K[[z_1, \dots, z_c]]/(H_1, \dots, H_c)$$

is generated by the nonzero class of $\det(\partial H_a / \partial z_b)$; see [22] and [14, Theorems 7.1 and A.5]. We need a relative version in which parameters on the reduced space may be lifted arbitrarily to the thickening.

The following lemma is used in the proof of Lemma 3.1 to show that the relative Jacobian has a nonzero socle class in each local fiber algebra. The perfect residue pairing then gives the nonzero coefficients in the relation on S_1 .

Lemma 2.1. *Let $B = \mathbb{C}[[s_1, \dots, s_r]]$, $R = B[[z_1, \dots, z_c]]$, and*

$$A = R/(H_1, \dots, H_c), \quad \Delta = \det \left(\frac{\partial H_a}{\partial z_b} \right) \in A,$$

where $r, c \geq 1$ and the H_a form a regular sequence. Assume that A is finite flat over B and that $A_{\text{red}} \simeq B$ as a B -algebra. If N is the nilradical of A , then

$$\text{Ann}_A(N) = B\Delta.$$

Moreover, if $\tau_1, \dots, \tau_r \in A$ reduce to a regular system of parameters of A/N , then the image of Δ generates the socle of $A/(\tau_1, \dots, \tau_r)$ and is nonzero.

Proof. Translate each z_b by its image in $A/N = B$. Then the kernel of $R \rightarrow A/N$ is $(z_1, \dots, z_c)$, and differentiation in the z_b with the s_i fixed is unchanged. Let K be the fraction field of B and put $\mathfrak{p} = (z_1, \dots, z_c) \subset R$. Since $\sqrt{(H_1, \dots, H_c)} = \mathfrak{p}$, the finite local K -algebra $A_K = A \otimes_B K$ identifies with $R_{\mathfrak{p}}/(H_1, \dots, H_c)$. Completing $R_{\mathfrak{p}}$ gives $K[[z_1, \dots, z_c]]$ and leaves its Artinian quotient A_K unchanged. This presents A_K as a zero-dimensional complete intersection with residue field K . The Jacobian theorem therefore gives $N_K \Delta = 0$. Since A is B -flat, $A \rightarrow A_K$ is injective, and hence $N\Delta = 0$ in A .

Put $M = \text{Ann}_A(N)$. The closed fiber $A/\mathfrak{m}_B A$ is an Artinian complete intersection and hence Gorenstein. Lift a linear functional defining its perfect multiplication pairing to a B -linear map $\ell : A \rightarrow B$. The A -linear map

$$A \longrightarrow \text{Hom}_B(A, B), \quad a \longmapsto (b \longmapsto \ell(ab)),$$

is an isomorphism modulo $\mathfrak{m}_B$ and hence an isomorphism between free B -modules of the same rank. Finite adjunction consequently gives an isomorphism of B -modules

$$M = \text{Hom}_A(B, A) \simeq \text{Hom}_A(B, \text{Hom}_B(A, B)) \simeq B.$$

The Jacobian theorem on the closed fiber gives $\Delta \notin \mathfrak{m}_B A$, and therefore $\Delta \notin \mathfrak{m}_B M$. As M is free of rank one over B , it follows that $M = B\Delta$.

Now let $B' = \mathbb{C}[[T_1, \dots, T_r]]$ act on A by $T_i \mapsto \tau_i$. The induced map $B' \rightarrow A/N$ is an isomorphism. Each quotient in the finite filtration by powers of N is finite over A/N , so A is finite over B' . The τ_i form a system of parameters in the Cohen–Macaulay ring A , and hence a regular sequence. Since B' is regular of dimension r and the closed fiber has dimension zero, miracle flatness makes A finite free over B' [23, Tags 00R4 and 00NX]. Choose lifts of the τ_i to R . Together with $H_1, \dots, H_c$, they form a system of parameters of the regular local ring R , hence a regular sequence [23, Tag 02JN]. Thus the closed fiber $A/(\tau_1, \dots, \tau_r)$ is again a zero-dimensional complete intersection, so $\text{Hom}_{B'}(A, B')$ is free of rank one over A by the same argument.

The quotient $A \rightarrow A/N = B'$ splits as a map of B' -modules. Dualizing gives a split injection $B' \hookrightarrow \text{Hom}_{B'}(A, B')$ whose image consists of the functionals vanishing on N . Under an A -linear identification $\text{Hom}_{B'}(A, B') \simeq A$, this image is precisely $\text{Ann}_A(N) = M$. Thus $M \hookrightarrow A$ splits over B' . The actions of B and B' on M both factor through A/N , so $M = B'\Delta$. Reduction modulo $(T_1, \dots, T_r)$ preserves the split injection. Hence Δ remains nonzero in $A/(\tau_1, \dots, \tau_r)$. The maximal ideal of this quotient is the image of N and annihilates Δ . Its socle is one-dimensional, proving the assertion. $\square$

3. RELATIONS ON A FIBER

Let $p_1, \dots, p_n$ be polynomials in affine coordinates $\xi_1, \dots, \xi_n$, with an isolated common zero x . Write the local Grothendieck residue as

$$\rho_x(P) = \text{Res}_x \left[\frac{P d\xi_1 \wedge \dots \wedge d\xi_n}{p_1 \cdots p_n} \right].$$

The pairing $(a, b) \mapsto \rho_x(ab)$ is perfect on the local complete-intersection algebra [7, Theorem 1.5.1]. If the highest homogeneous parts of the p_i have no common nonzero zero, Euler–Jacobi vanishing gives

$$(3.1) \quad \sum_x \rho_x(P) = 0 \quad \text{for} \quad \deg P < \sum_{i=1}^n (\deg p_i - 1).$$

The sum includes all common zeros, and the fiber algebras need not be reduced [7, Theorem 1.5.12].

The following lemma supplies the relation on S_1 used in Lemma 4.2. That lemma turns the relation into a homogeneous polynomial that is nonzero on V and vanishes on the other fibers.

Lemma 3.1. *In Setup 1.4, there are nonzero scalars λ_x , $x \in S_1$, such that, for every homogeneous polynomial P of degree t ,*

$$\sum_{x \in S_1} \lambda_x P(x) = 0.$$

Proof. Put $y = y_1$ and $c = n - r$. Choose target coordinates $[Y_0 : \dots : Y_n]$ with $y = [1 : 0 : \dots : 0]$, and write $u_i = Y_i/Y_0$. We may assume that

$u_1, \dots, u_r$ restrict to parameters on V at y , while $du_{r+1}, \dots, du_n$ span its conormal space there. The completed ideal of V at y then has generators

$$h_a = u_{r+a} + O((u_1, \dots, u_n)^2), \quad 1 \leq a \leq c.$$

In particular, $u_1, \dots, u_r, h_1, \dots, h_c$ are ambient local parameters.

Write $f = [f_0 : \dots : f_n]$, where the f_i are degree- q forms without a common zero. Choose a source hyperplane avoiding $f^{-1}(y)$ and not contained in any $\{f_i = 0\}$ for $1 \leq i \leq n$. On its affine complement, let p_i be the dehomogenization of f_i , for $0 \leq i \leq n$. Then $\deg p_i = q$ for $1 \leq i \leq n$, and $p_1 = \dots = p_n = 0$ defines the whole ambient fiber. Their highest homogeneous parts have no common nonzero zero: such a zero would give a point at infinity with $f_1 = \dots = f_n = 0$ and $f_0 \neq 0$, contrary to the choice of the hyperplane.

Choose general affine linear coordinates $(s_1, \dots, s_r, z_1, \dots, z_c)$ so that the ds_i restrict to a cotangent basis of V at every point of S_1 . On this affine chart, define the polynomial

$$\Theta = \det \left(\frac{\partial p_{r+a}}{\partial z_b} \right)_{1 \leq a, b \leq c}.$$

It has degree at most $c(q-1)$.

Claim 3.2. *For every $x \in S_1$,*

$$\text{Soc}(\mathcal{O}_{\mathbf{P}^n, x}/(p_1, \dots, p_n)) = \mathbb{C} \cdot \bar{\Theta} \neq \{0\}.$$

Proof of Claim 3.2. Fix $x \in S_1$ and center the source coordinates at x . In the completed local ring, V is a graph $z_b = \varphi_b(s)$. Set $w_b = z_b - \varphi_b(s)$, so that differentiation in w_b agrees with differentiation in z_b when s is fixed. The function p_0 is a unit at x . Put

$$v_i = p_i/p_0 = u_i \circ f, \quad H_a = h_a(v_1, \dots, v_n), \quad A = \widehat{\mathcal{O}}_{\mathbf{P}^n, x}/(H_1, \dots, H_c).$$

Total invariance gives $\sqrt{J} = I$ for the local ideals I of V and J of $f^{-1}(V)$. Thus $I^e \subseteq J \subseteq I$ for some $e \geq 1$, and after completion

$$(w_1, \dots, w_c)^e \subseteq (H_1, \dots, H_c) \subseteq (w_1, \dots, w_c).$$

Hence $A_{\text{red}} = B = \mathbb{C}[[s_1, \dots, s_r]]$, and A is finite over B , generated by the monomials in w of total degree less than e . Since $(H_1, \dots, H_c)$ has height c , it is generated by a regular sequence, so A is Cohen–Macaulay [23, Tag 02JN]. Miracle flatness then makes A finite free over B [23, Tags 00R4 and 00NX].

Since g is étale at x , the reductions of $v_1, \dots, v_r$ are parameters of A_{red} . Apply Lemma 2.1 with normal variables w_b and parameter lifts $\tau_i = v_i$. It follows that the relative Jacobian

$$\Delta = \det \left(\frac{\partial H_a}{\partial w_b} \right) = \det \left(\frac{\partial H_a}{\partial z_b} \right)$$

has nonzero socle class in $A/(v_1, \dots, v_r)$. Since $u_1, \dots, u_r, h_1, \dots, h_c$ are local parameters at y , this quotient identifies with the local fiber algebra,

namely the completed source ring modulo $(p_1, \dots, p_n)$. In this quotient the chain rule gives

$$\frac{\partial H_a}{\partial z_b} \equiv \frac{\partial v_{r+a}}{\partial z_b} \equiv p_0^{-1} \frac{\partial p_{r+a}}{\partial z_b}.$$

Here $H_a - v_{r+a} \in (v_1, \dots, v_n)^2$, and terms divisible by the p_i vanish in the fiber algebra. Taking determinants yields

$$\Delta \equiv p_0^{-c} \Theta \pmod{(p_1, \dots, p_n)}.$$

Since p_0 is a unit and the Artinian local fiber algebra equals its completion, Θ has nonzero socle class. $\square$

Using the fixed affine coordinates (s, z) , set $\lambda_x = \rho_x(\Theta)$. By Claim 3.2,

$$\rho_x(\Theta P) = \lambda_x P(x) \quad \text{for every local function } P.$$

Moreover, the nonzero socle class in Claim 3.2 and the perfect residue pairing imply $\lambda_x \neq 0$.

Since $c = n - r$ and $t = r(q - 1) - 1$, a polynomial P of degree at most t satisfies

$$\deg(\Theta P) \leq c(q - 1) + t = n(q - 1) - 1.$$

The support of the ambient fiber is exactly S_1 , so (3.1) gives

$$0 = \sum_{x \in S_1} \rho_x(\Theta P) = \sum_{x \in S_1} \lambda_x P(x).$$

Let L_0 be a linear form defining the chosen source hyperplane, and let v_x be the fixed homogeneous coordinate representative of x . Every homogeneous polynomial P of degree t dehomogenizes to a polynomial of degree at most t , with value $P(v_x)/L_0(v_x)^t$ at x . Replacing each weight λ_x by $\lambda_x/L_0(v_x)^t$ gives the required homogeneous relation. $\square$

4. LIFTING RELATIONS BY DUALITY

Let $\omega_V = \mathcal{H}^{-r}(\omega_V^\bullet)$ be the canonical sheaf of V , where $\omega_V^\bullet$ is its normalized dualizing complex. It is coherent and torsion-free of rank one and agrees with the canonical line bundle on V_{reg} [23, Tags 0FVV and 0AXY].

The adjunction twist $rq - t = r + 1$ is independent of q . The following lemma embeds $\omega_V(r + 1)$ into a fixed twist of $\mathcal{O}_V$.

Lemma 4.1. *Let $V \subseteq \mathbf{P}^n$ be integral of dimension r . There is an integer $m > 0$, depending only on this embedding, and an injection*

$$\omega_V(r + 1) \hookrightarrow \mathcal{O}_V(m).$$

Proof. For $m \gg 0$, the sheaf $\mathcal{H}om_V(\omega_V(r + 1), \mathcal{O}_V)(m)$ is globally generated. It has rank one, so a section nonzero at the generic point gives a generically injective morphism $\omega_V(r + 1) \rightarrow \mathcal{O}_V(m)$. The kernel is torsion and hence zero, since ω_V , and hence $\omega_V(r + 1)$, is torsion-free. $\square$

This is the final lemma required for the proof of [Linearity Theorem](#). It turns the relation on S_1 into a homogeneous polynomial of fixed degree

vanishing on the other fibers in Setup 1.4. Under iteration, S and Q may vary, but Q can always be chosen of the same degree m .

Lemma 4.2. *There is an integer $m > 0$, depending only on the embedding $V \subseteq \mathbf{P}^n$, such that for every choice of data in Setup 1.4 with $q > md$, there exists a homogeneous polynomial Q of degree m satisfying*

$$Q|_V \neq 0, \quad Q|_{S \setminus S_1} = 0.$$

Proof. By Lemma 4.1 and Serre vanishing [18, Theorem III.5.2], fix $m > 0$ with an injection $\omega_V(r+1) \hookrightarrow \mathcal{O}_V(m)$ such that every section of $\mathcal{O}_V(m)$ extends to a homogeneous polynomial of degree m . Also choose $N \geq 1$ such that

$$H^1(\mathbf{P}^n, \mathcal{I}_V(s)) = 0, \quad H^i(V, \mathcal{O}_V(s)) = 0 \quad (i > 0)$$

for all $s \geq N$ (in fact, both vanishings hold for every $s > 0$ by [9, Proposition 2.3] and the standard cohomology sequence). Enlarging m if necessary, retain the injection and extension property above and arrange that $md \geq N + r + 1$. Then $q > md$ ensures the vanishing conditions used below. These choices depend only on the embedding of V .

By Lemma 3.1, choose nonzero λ_x , $x \in S_1$, annihilating every homogeneous polynomial of degree t . Put $\sigma_i = \ell_i|_V$ and extend the λ_x by zero on $S \setminus S_1$. Using the chosen coordinate representatives to evaluate degree- t sections on S , the weights define a nonzero functional λ on $H^0(S, \mathcal{O}_S(t))$. Since $t \geq N$, every section of $\mathcal{O}_V(t)$ extends to a degree- t homogeneous polynomial. Thus λ annihilates the restrictions of all sections of $\mathcal{O}_V(t)$.

Write $i : S \hookrightarrow V$ and set $E = i_*\mathcal{O}_S(t)$. For $0 \leq j \leq r$, put

$$K_j = \mathcal{O}_V(t - jq)^{\oplus \binom{r}{j}}.$$

The ordered sections $\sigma_1, \dots, \sigma_r$ give the twisted Koszul resolution

$$(4.1) \quad 0 \longrightarrow K_r \longrightarrow \dots \longrightarrow K_0 \longrightarrow E \longrightarrow 0.$$

Indeed, at a point of S they generate the maximal ideal of a regular local ring of dimension r ; away from S one of them is a unit in a local trivialization. The leftmost term is

$$M = K_r = \mathcal{O}_V(t - rq) = \mathcal{O}_V(-r - 1).$$

All other terms have sufficiently large twists, since

$$t - jq = (r - j)q - r - 1 \geq q - r - 1 \geq N \quad (0 \leq j < r).$$

Their higher cohomology vanishes by the choice of N . The successive connecting maps therefore give an exact sequence

$$(4.2) \quad H^0(V, \mathcal{O}_V(t)) \longrightarrow H^0(S, \mathcal{O}_S(t)) \xrightarrow{\partial} H^r(V, M) \longrightarrow 0.$$

We show that, under duality, the transpose of ∂ is restriction of sections of $\omega_V(r+1)$ to S . Let

$$\varepsilon \in \text{Ext}_V^r(E, M)$$

be the extension class of (4.1); its connecting homomorphism is ∂ . Proper duality gives

$$(4.3) \quad \begin{aligned} H^0(V, \omega_V(r+1)) &\simeq \operatorname{Hom}_V(M, \omega_V) \simeq H^r(V, M)^\vee, \\ \operatorname{Ext}_V^r(E, \omega_V) &\simeq H^0(S, \mathcal{O}_S(t))^\vee. \end{aligned}$$

Here $\vee$ denotes the vector-space dual. We regard a section a of $\omega_V(r+1)$ as a morphism $a : M \rightarrow \omega_V$ via the first identification. The first line uses top-degree duality. For the second line, the canonical morphism $\omega_V[r] \rightarrow \omega_V^\bullet$ is an isomorphism near S . As E is supported on S , applying $R\mathcal{H}om_V(E, -)$ gives an isomorphism, and proper duality applies. Both duality identifications are compatible with morphisms and connecting maps [23, Tag 0FVV(5),(6)]. Hence $\partial^\vee$ is, up to the common sign convention, the map

$$\operatorname{Hom}_V(M, \omega_V) \longrightarrow \operatorname{Ext}_V^r(E, \omega_V), \quad a \longmapsto a_*\varepsilon,$$

obtained by pushing the extension along $a : M \rightarrow \omega_V$.

To compute this map, apply $\mathcal{H}om_V(-, \omega_V)$ to (4.1). Its final differential fits into the exact sequence

$$\omega_V(r+1-q) \oplus^r \xrightarrow{(\pm\sigma_1, \dots, \pm\sigma_r)} \omega_V(r+1) \longrightarrow i_*(\omega_V(r+1)|_S) \longrightarrow 0.$$

There are no other cohomology sheaves: near S this is the dual Koszul complex of a regular sequence on an invertible sheaf, and away from S the complex is contractible. It follows that

$$(4.4) \quad \operatorname{Ext}_V^r(E, \omega_V) \simeq H^0(S, \omega_V(r+1)|_S).$$

The class $a_*\varepsilon$ is represented by $a : K_r = M \rightarrow \omega_V$ in the last term of the dual complex, so its image is precisely $a|_S$. Dualizing (4.2) therefore lifts λ to a section $a \in H^0(V, \omega_V(r+1))$.

Put $L_x = \mathcal{O}_S(t)|_x$ and $W_x = \omega_V(r+1)|_x$ for $x \in S$. The isomorphism (4.4) and the second line of (4.3), with the common sign convention fixed, induce

$$\beta = \bigoplus_{x \in S} \beta_x : \bigoplus_{x \in S} W_x \xrightarrow{\sim} \left(\bigoplus_{x \in S} L_x \right)^\vee, \quad \beta_x : W_x \xrightarrow{\sim} L_x^\vee.$$

These maps respect the decomposition of E over the points of S . Using the coordinate representative v_x to define the evaluation functional $\operatorname{ev}_{v_x} \in L_x^\vee$, the lifted section satisfies

$$\beta_x(a|_x) = \lambda_x \operatorname{ev}_{v_x}, \quad a|_x = 0 \iff \lambda_x = 0.$$

Since $\lambda \neq 0$ and is supported on S_1 , we obtain

$$a \neq 0, \quad a|_{S \setminus S_1} = 0.$$

Let $\iota : \omega_V(r+1) \hookrightarrow \mathcal{O}_V(m)$ be the fixed injection. Lift $\iota(a)$ to a homogeneous polynomial Q of degree m . Then

$$Q|_V = \iota(a) \neq 0, \quad Q|_{S \setminus S_1} = 0,$$

as required. $\square$

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